



Design thinking and visualizations

Hey, welcome back. We've covered a lot of ground in our exploration of data visualizations. We've talked a lot about how your audience should be the focus when you are making decisions about charts, colors, space, labels and everything else that goes into a data viz. Now let's talk about design thinking. Design thinking is a process used to solve complex problems in a user-centric way. When you bring design thinking into your work, you're trying to identify alternative strategies for your visualizations that might not be clear right away. You have to challenge your own thinking and explore different ways of approaching the problems and finding solutions.

Airbnb is one example of a company that use a design thinking approach to help their business grow. When the company a vacation rental online marketplace, wasn't generating as much revenue as they wanted, they decided to start experimenting. Even though the data they collected and analyzed was valuable, they needed to look at their product through the eyes of the customer. They realized the photos of the places that customers were seeing just weren't very good, so they decided to help their customers replace the not so great photos with more professional looking ones. So they hired a photographer and went door to door to take professional photos of their New York City listings. In a week, the listings with these photos saw 2 to 3 times more bookings, and their revenue nearly doubled, thanks to their new design thinking, user based mindset. If design thinking can work for companies like Airbnb, it can help data analysts too, and data visualization is the perfect stage of your analysis to apply a user based mindset. If you use design thinking when planning and creating your data viz, you'll be making decisions based on the needs of the people who will be viewing them. This way your audience will be engaged and enlightened by how you visualize your findings. While the design thinking process comes in lots of different forms, they all have stages or phases.

We'll talk about **five phases** that you can use when creating data visualizations, **empathize, define, ideate, prototype, and test**. In the spirit of design thinking, these phases don't have to follow a set order. Instead, think of them as an overview of actions that can help you produce a user centered design in your visualizations. In the empathize phase you think about the emotions and needs of the target audience of your data viz, whether it's stakeholders, team members or the general public.

Here you should avoid areas where people might face obstacles interacting with your visualizations. For example, let's say you've been working on an analysis for a pharmaceutical company about how patients have been responding to a new treatment. You're getting ready to visualize the data, so you should think about the audience, which will include stakeholders like pharmacists, doctors and other medical professionals. Maybe you're thinking of using a color scheme that you like, but you realize that these colors might be a challenge to some people. The colors might be too bright or dramatic, which might not be right for the seriousness of the data. Or the colors might not have enough contrast for people who have color vision deficiencies. By adjusting the colors, you'll be empathizing with the needs of your audience.

If there's someone on your team who is vision impaired, you want to find a way to explain the data verbally as well. The define phase helps you to find your audiences needs, their problems, and your insights. This goes hand in hand with the empathize phase as you'll use what you learned in that phase to help you spell out exactly what your audience needs from your visualization. You could use this phase to think about which data to show in your visualization. Maybe this data viz will also be presented to patients who are part of your company's study. While you'll need to meet your objectives, there might be data that could make these people uncomfortable. You can think of ways to position that data to make it more digestible.

Or if you're presenting to different audiences, you can adjust your visualizations to meet each group's needs by seeking input from members of the group or colleagues who've worked with that group before. In the ideate phase, you start to generate your data viz ideas. You'll use all of your findings from the empathize and define phases to brainstorm potential data viz solutions. This might involve creating drafts of your visualization with different color combinations or maybe experimenting with different shapes. Creating as

many examples as possible will help you refine your ideas. The key here is to always remember your audience when coming up with ideas and strategies. You want to think about how you can position your visualizations to meet the needs and expectations of your audience.

The final two phases are prototype and test. Here you'll start putting your charts, dashboards or other visualizations together. If you've kept your audience in mind through all the phases to this point, then your data viz will be informative and approachable. You might want to create lots of visualizations to choose which one best meets your objective. You could test your visualizations by showing them to team members before presenting them to stakeholders. If you've created more than one for the same data, or for different audiences like the medical professionals and the patients from our earlier example, you can share all of your options. As always, listen to any feedback you get.

Critiques both your own and others are key to the design thinking process. They help you keep your focus on the audience by integrating new ideas in your final product. The phrase thinking outside the box is used a lot, but it definitely applies here. The box in this case is your own usual way of approaching data, and its visualization. If you embrace design thinking, you'll be able to create super effective data viz for any audience. Up next, we'll cover more things you need to consider within your data viz. See you there.